

CMS - Surgery - Form 2

Answers

1. **D** BAD QUESTION. Explanation is that severe thrombocytopenia + local skin necrosis = Dx of HIT. This is an old question so D might be the right answer here, but this question isn't compatible with the new guidelines of HIT. We use the 4T score to assess for HIT, Thrombocytopenia, Thrombosis, Timing and Other causes more likely. The timing itself is off as HIT occurs within 5-10 days and carries a 95% sensitivity and a 97% specificity for that timeframe. Secondly, if we look at the case clinically, we see multiple ecchymoses along with a cyanotic toe. A cyanotic toe is a thromboembolic phenomenon, the ecchymoses aren't - that's livedo reticularis. Given the presentation, the cause of livedo reticularis is cholesterol embolism. Lastly, Heparin doesn't reduce the PTT time, it increases it. Besides, what role does Heparin play in a CABG patient?
2. **C**. Duodenal hematomas most commonly occur following blunt abdominal trauma and are more commonly seen in children due to minor anatomic differences including thinner abdominal wall musculature and less adipose tissue. They generally occur when a blunt force rapidly compresses the duodenum against the vertebral column. Blood collects between the submucosal and muscular layers of the duodenum causing partial or complete obstruction. Patients classically present 24-36 hours after the trauma with epigastric pain and bilious vomiting. Diagnosis is confirmed by CT. Most will resolve in 1-2 weeks. Management involves decompression by NGT and often with parenteral nutrition. Surgery or percutaneous drainage may be considered to evacuate the hematoma if nonoperative management fails.
3. **E**. Surgical excision of the cyst is indicated. Choledochal cysts (a.k.a. bile duct cyst) are congenital conditions involving cystic dilatation of bile ducts. They are uncommon in western countries but not as rare in East Asian nations like Japan and China. Most patients have symptoms in the first year of life. It is less common (20% of cases) for symptoms to be undetected until adulthood, and usually adults have associated complications. The classic triad of intermittent abdominal pain, jaundice, and a right upper quadrant abdominal mass is found only in a minority of patients. In infants, choledochal cysts usually lead to obstruction of the bile ducts and retention of bile. This leads to jaundice and an enlarged liver. If the obstruction is not relieved, permanent damage may occur to the liver (scarring and cirrhosis) with signs of portal hypertension (due to obstruction of blood flow through the liver) and ascites (fluid accumulation in the abdomen). There is an increased risk of cancer in the wall of the cyst. In older individuals, choledochal cysts are more likely to cause abdominal pain and intermittent episodes of jaundice and occasionally cholangitis (inflammation within the bile ducts caused by the spread of bacteria from the intestine into the bile ducts). Pancreatitis also may occur. The cause of these complications may be related to either abnormal flow of bile within the ducts or the presence of gallstones. They are classified into 5 types but that information is not contained here. The surgical treatment of choledochal cysts depends on the type of cyst. This is a "type 1" meaning that it is a cystic dilation of the common duct, which is treated with cystectomy > PCT drainage. For "type 4 (or 5)" meaning that there is cystic dilation of the intrahepatic ducts, the treatment is transplant +/- Kasai procedure).
4. **E**. This patient likely has rupture of the thoracic aorta. Patients with rapid deceleration injury can have aortic injury which can lead to sudden death due to aortic rupture and exsanguination. Very few patients will have an incomplete or contained rupture. Hypotension (or hypertension), external trauma, and altered mental status are often found. These patients require chest x-ray, CT, and possibly transesophageal echo as well. Radiographic findings may include widened mediastinum, large left-sided hemothorax, deviation of the mediastinum to the right, and disruption of the normal aortic contour. Management includes antihypertensive therapy and immediate operative repair.
5. **E**. Thyroidectomy is indicated as FNA is not reliable for follicular cancers due to the high probability of false negatives and because FNA cytology cannot distinguish between follicular adenoma and carcinoma since histology showing capsular invasion is required. Thus thyroidectomy is indicated for follicular cytology.

6. **C.** The barium x-ray shows a stricture in the terminal ileum, which is the most commonly affected area in Crohn's disease. Crohn's causes serpiginous ulceration leading to strictures, which is what has happened here. Colon cancer may also cause strictures (the "string sign"), but this stricture is not in the colon. His age as well as his history of mild abdominal pain and diarrhea along with his weight loss and low grade temperature elevation also point to Crohn's.
7. **E.** A Dipyridamole-Thallium scan is a myocardial perfusion study; it looks at blood supply to the heart muscle while under the relative stress of dipyridamole, which causes the coronary arteries to slightly and temporarily constrict (become narrower). Thallium is the radio-isotope used to 'light up' the blood cells. This patient has an abdominal aortic aneurysm and cannot undergo the typical stress tests. Preoperative planar dipyridamole-thallium scanning (DTS) is most useful to stratify selected nonvascular surgery patients at intermediate or high risk by clinical assessment.
8. **E.** Venous valvular insufficiency leads to pooling. Previous leg injury increases risk. Venous stasis ulcers are classically found above the medial malleolus.
9. **F.** Retrograde urethrography is faster and has largely replaced IVP (choice E) for diagnosis of urethral injury.
10. **E.** This patient has a varicocele. Dilated veins in the pampiniform plexus occur as a result of increased venous pressure, usually on the left (due to increased resistance from left gonadal vein drainage). If there is a thrombus or a tumor on the right, treat with varicolectomy, if on the left, treat with embolization by interventional radiology. The stasis raises the scrotal temperature causing damage to sperm, which require slightly cooler temperatures.
11. **B.** Indomethacin closes the PDA via cyclooxygenase inhibition with increased norepinephrine release. Indomethacin is a potent prostaglandin inhibitor and is the mainstay treatment to close a patent ductus arteriosus. Prostaglandin E1 acts as a vasodilator and can prevent the PDA from closing if necessary (e.g. if required for life due to another congenital abnormality such as transposition of the great vessels).
12. **E.** Secondary hyperparathyroidism occurs in renal failure when the kidneys both cannot excrete sufficient phosphate and cannot produce enough vitamin D. Both lead to decreased serum calcium, triggering increased PTH release from the parathyroid glands. Untreated patients present with bone demineralization, calcification of soft tissues, accelerated vascular calcification, pruritus, and painful skin ulcerations from subcutaneous Ca^{2+} deposits. Remember, in primary hyperparathyroidism, calcium is and PTH are high (calcium is high because PTH is being produced excessively and without appropriate regulation); in secondary hyperparathyroidism, calcium is low and PTH is high (PTH is high because calcium is low).
13. **A.** This woman has choledocholithiasis, likely due to an undiagnosed common bile duct stone. Negatives that are obviously pointed out in a stem are there for a reason. The stem says intra-operative cholangiography was not performed. There is such a thing as primary biliary cirrhosis/cholangitis, but primary biliary sclerosis (choice E) does not exist.
14. **E.** This is ovarian torsion until proven otherwise. Exploratory laparotomy is the next step. Management of a ruptured cyst would be admission and observation as most have an uncomplicated course, but if the cyst had ruptured, fluid would have been detected and the mass would not remain 10 cm.
15. **E.** Normal Anion Gap has two major differentials: GI loss and renal tubular acidosis. This patient had a pancreaticoduodenectomy (Whipple procedure) connecting the pancreas directly to the GI tract, which has resulted in loss of bicarbonate and elevated amylase as one of the specific complications of this procedure is an anastomotic leak.
16. **E.** This patient has a toxic thyroid adenoma as indicated by her hyperthyroid state and solitary nodule with focal RAUI uptake (Graves' disease produces diffuse RAUI uptake). Treatment is lobectomy with administration of beta blocker and antithyroid drugs to induce euthyroid state prior to surgery.
17. **E.** This patient has Fournier's gangrene, treatment is urgent (minutes count), surgical debridement is indicated.

18. A.

Table 2. Onset, Maximum Dose, and Duration of Anesthetics

Drug	Onset	Maximum Dose (With Epinephrine)	Duration (With Epinephrine)
Lidocaine	Rapid	4.5 mg/kg (7 mg/kg)	120 min (240 min)
Mepivacaine	Rapid	5 mg/kg (7 mg/kg)	180 min (360 min)
Bupivacaine	Slow	2.5 mg/kg (3 mg/kg)	4 hours (8 h)
Ropivacaine	Medium	2-3 mg/kg	3 hours (6 h)

19. C. This is a benign lesion found in the tail of the pancreas. There is no need to take out the entire pancreas if you can avoid it. A distal pancreatectomy will get rid of the mass and retain a functioning pancreas.

20. E. The inflammatory cytokines involved in wasting diseases are interleukin 6, TNF-alpha, IL1B, and interferon-gamma. TNF-alpha has been shown to have direct catabolic effect on skeletal muscle and adipose tissue and produces muscle atrophy through the ubiquitin–proteasome proteolytic pathway. The mechanism involves the formation of reactive oxygen species leading to upregulation of the transcription factor NF-κB. NF-κB is a known regulator of the genes that encode cytokines, and cytokine receptors. The increased production of cytokines induces proteolysis and breakdown of myofibrillar proteins.

21. A. Cancel the operation. There is no reason to believe that this patient is not competent. Fear is all the reason she needs. Autonomy must be respected.

22. B. HTN is a likely cause for intraparenchymal hemorrhage, along with amyloid deposition and tumors.

23. D. Lower GI bleed in a patient 40 or older is cancer until proven otherwise even in the presence of other known causes of bleeding such as internal hemorrhoids. Do a colonoscopy to rule out cancer.

24. B. No trachea shift means Choice E is out. Choice A would not completely do away with all breath sounds on the right. Pulmonary embolism wouldn't do away with breath sounds at all at first and is usually the result of a DVT and not of a puncture wound. Ruptured bronchus would cause crepitus and lots of other things not seen here. The low BP suggests hemorrhage. Hemothorax may or may not cause tracheal deviation. Even when it does, it isn't usually significant. This is because unlike air, it doesn't cause the "tension" that tension pneumothorax causes.

25. H. Tenderness to palpation along the medial joint line with effusion should raise suspicion for a meniscal tear. Getting up from a low position is a classic presentation. Ligament stability is normal which rules out Choices A, D, F, and G.

26. C. The diagnosis of acute diverticulitis should be suspected in a patient with lower abdominal pain and abdominal tenderness on physical examination. The pain is usually in the left lower quadrant in Western populations but may be suprapubic or in the right lower quadrant, particularly in Asians. Laboratory findings of leukocytosis, while not sensitive or specific for acute diverticulitis, can support the diagnosis. Imaging (preferably computed tomography scan) is required to establish the diagnosis of acute diverticulitis.

27. **G.** Vesicular lesions in eye make herpes simplex conjunctivitis the likely diagnosis.
28. **A.** Central retinal artery occlusion presents with acute painless monocular vision loss. The retina is cloudy with attenuated vessels and cherry red spot at fovea, in contrast to central vein occlusion which presents as retinal hemorrhage.
29. **D.** The tumor has likely returned. Maintaining an endotracheal airway will be very problematic. A tracheostomy would be below the lesion and maintain airway.
30. **E.** Nerve conduction study is first line for evaluating carpal tunnel syndrome. EMG would be supporting but not definitively confirm the diagnosis as the muscle wasting could theoretically be from another cause.
31. **E.** Pharyngoesophageal (Zenker) diverticulum commonly presents with halitosis and recurrent aspiration pneumonia.
32. **E.** This patient has a ruptured diaphragm with the GI tract where it shouldn't be well above the level of the right diaphragm.
33. **C.** Insulinoma produces many of the same symptoms as hypoglycemia. Insulinomas are tumors of the pancreas that produce excessive amounts of insulin leading to low blood sugar (<50). They are more common in women, are usually small (<2 cm), and more than 90% are benign (non-cancerous).
34. **B.** This patient has cholecystoduodenal fistula with an impacted gallstone, i.e. gallstone ileus. Pneumobilia usually can be distinguished from air in the portal venous system by its appearance on computed tomography (CT) scan. The most common conditions associated with pneumobilia include a biliary-enteric surgical anastomosis, an incompetent sphincter of Oddi, or a spontaneous biliary-enteric fistula, which is what we see here. The stone has become impacted at the ileocecal valve, resulting in the obstruction.
35. **E.** Esophagogastroduodenoscopy is indicated. When solids get "stuck" more so than liquids, we should look for cancer. When liquids are more difficult to swallow than solids, look for other etiologies. This patient has several risk factors for squamous cell cancer of the esophagus and EGD to investigate/biopsy is warranted.
36. **B.** Very high WBC count in the presence of silent bowel is very concerning for acute mesenteric ischemia. A patient with a history of AFIB is likely to throw an embolism occluding the SMA. The metabolic acidosis is consistent with this theory. Exploratory celiotomy (i.e. exploratory laparotomy) is warranted immediately.
37. **D.** Repeat typing and crossmatching of the transfused blood is indicated. The patient is having an acute hemolytic reaction to the infused blood products due to ABO incompatibility. A Coombs test will confirm diagnosis.
38. **D.** Mid-systolic click at the apex indicates mitral valve prolapse (MVP). The rest of the stem is just supporting this diagnosis.
39. **A.** Antibiotic coverage for asplenic patients with sepsis should cover encapsulated bacteria (Ceftriaxone). Vancomycin should be administered for all the regular players as well. Her immunizations need updated.
40. **B REMOVED QUESTION.** Carotid duplex ultrasonography is indicated to confirm TIA from carotid artery atherosclerosis.
41. **D.** Urine output is one of the best indications of adequate resuscitation. The average male should put out a total of 0.5 to 1.0 mL of urine per kg per hour. For a 70 kg male that will be 35-70 mL/hr. Pulse is not as reliable because it may be elevated for reasons other than hypovolemia.

42. **A.** Obesity is a risk for cholecystitis. Remember the F's: Female, Fat, Forties, Fertile. This is an obese female.
43. **C.** Intubation and hyperventilation are indicated. Remember the rhyme: If GCS is less than 8, first thing you do is intubate. Hyperventilate to decrease ICP via intracerebral vasoconstriction due to decreased PaCO_2 .
44. **A** BAD QUESTION (based on outdated information). The pathogenesis of hidradenitis suppurativa is not fully understood. However, an explosion of research has led to greater understanding of the processes involved. As opposed to early theories that implicated apocrine glands as the primary contributors to HS/AI, most authors now support follicle-centered theories for the pathogenesis of HS/AI. Follicular occlusion, follicular rupture, and an associated immune response appear to be important events in the development of the clinical manifestations of HS/AI (UTD).
45. **E.** This patient has a Cushing ulcer from increased intracranial pressure. This is caused by direct stimulation of vagal nuclei from increased intracranial pressure (cerebral edema in this patient). Since it seems like the ulcer has perforated (free air under the diaphragm), laparotomy is indicated.
46. **A.** This is a basal cell carcinoma. "Pearly" is usually enough to suspect basal cell. Squamous cell is usually seen below the mouth (classically on the bottom lip), while basal cell is seen on the head at the level above the mouth. Mnemonic: Basal Nasal, Squamous Squats (lower lip)
47. **A.** Observation is warranted. Tall, skinny boys are particularly susceptible to spontaneous pneumothorax. But it typically resolves on its own in 3-5 days.
48. **B.** This patient likely has hypovolemia due to dehydration as she was found 48 hours after her accident. Signs of hypovolemia include hypotension, tachycardia, diaphoresis, cool, clammy extremities, decreased right atrial pressure and PCWP, increased SVR, decreased cardiac output (due to loss of circulating blood volume from hemorrhage or interstitial from bowel obstruction), excessive vomiting or diarrhea, and polyuria.
49. **C.** This patient likely has collapse of the right middle lobe of the lung from decreased inspiratory effort. Atelectasis is a complication most often found in the first post-operative day. The x-ray is consistent with this diagnosis as well.
50. **H.** This patient has developed toxic megacolon (diameter >6 cm) as a complication of ulcerative colitis, which is one of the few diseases that can cause diarrhea for 6 weeks in an immunocompetent patient.

In next pages, you will find the answers made by someone else but the order of answers is not same. I have left it there in case it helps you better understand the answers.

NBME Surgery Form 2 - Answers & Explanations

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1) Toxic thyroid adenoma

- Focal patch of autonomously functioning thyroid tissue = “hot nodule”

2) Nerve conduction studies

- Confirmatory study for carpal tunnel syndrome (shows slowing in median nerve at wrist)
- EMG rules out other things such as polyneuropathy, plexopathy, radiculopathy, etc.

3) Loss of bicarbonate from the gastrointestinal tract

- Drainage of pancreatic fluid can cause metabolic acidosis from loss of bicarbonate

4) CT scan of the abdomen and pelvis

- Acute abdominal pain in LLQ, fever, leukocytosis = acute diverticulitis
- CT is diagnostic
- Tx: NPO, IVF, Abx

5) Duodenal hematoma

- Contrast CT to confirm the diagnosis
- Hematoma may obstruct duodenal lumen → double bubble sign
- Diagnosis may be delayed as the retroperitoneal location prevents peritonitis

6) Urine output of 30-40 mL/h

- Adequate end-organ perfusion is best indicated by urine output of 0.5-1 mL/kg/h; average adult male weighs ~70kg (70 x 0.5 = 35 mL/h)
- HR, mental status, and cap refill may be affected by underlying disease processes (less reliable markers)

7) Pharyngoesophageal (Zenker) diverticulum

- Presents with dysphagia, obstruction, halitosis, and aspiration (leading to multiple episodes of pneumonia)
- Common in elder males

8) Esophagogastroduodenoscopy

- Dysphagia to solids, age >50, chronic smoker/drinker = concern for esophageal cancer
- Definitive diagnosis of esophageal cancer = esophageal endoscopy with biopsy
- Subsequent CT scan can be used for staging or if endoscopy is unrevealing

9) Laparotomy

- perforated bowel → free air under diaphragm → laparotomy

10) Cholecystoduodenal fistula with an impacted gallstone

- Multiple dilated loops of small bowel = obstruction
- Gas within small bowel lumen and liver suggests communication (fistula) between enteric and biliary tracts
- Stone leaves through fistula to cause sx's

11) Distal pancreatectomy

- Large (>4 cm) cystadenomas should be resected

12) Venous valvular insufficiency

- Ulcer on medial malleolus
- Post-thrombotic syndrome = development of chronic venous insufficiency following DVT

13) Repeat typing and crossmatching of the transfused blood

- Crossmatching error - the patient is having an acute hemolytic reaction, most likely due to ABO incompatibility

14) Hypertension

- Post-op HTN associated with cerebral hyperperfusion syndrome - manipulation of the carotid artery/sinuses lead to baroreceptor dysfunction

15) Massive hemothorax

- Hypotension + flat neck veins + absent breath sounds after knife wound = hemothorax

16) Crohn disease

- “string sign” on barium x-ray = thin luminal contrast, usually in terminal ileum from spasm and eventual fibrosis

17) Colonoscopy

- Age >50 with lower GI bleed → colonoscopy
- In all anorectal problems, cancer has to be ruled out first; once diagnosis has been confirmed, internal hemorrhoids are tx with rubber-band ligation, whereas external hemorrhoids require surgery

18) Surgical excision of the cyst

- Choledochal cyst (cystic dilation of biliary tree)
- U/S is best non-invasive test; ERCP for definitive diagnosis
- Tx: surgical excision to relieve obstruction and biliary-enteric anastomosis
- Excision helps prevent malignant transformation (cholangiocarcinoma)

19) Hypovolemia

- Patient was down, laying on the floor for 48 hours (likely volume depleted)
- She hasn't moved → muscle breakdown and rhabdo → lactic acidosis → compensatory hyperventilation

20) Intubation and hyperventilation

- GCS <8 → Intubate
- Hyperventilation → ↓ CO₂ → vasoconstriction and ↓ ICP (CO₂ is a potent vasodilator in the brain)

21) Herpes simplex conjunctivitis

- Acute onset pain, conjunctival injection, vesicular lesions, visual blurring, discharge = HSV keratitis

22) Central retinal artery occlusion

- Acute painless monocular vision loss with pallor of optic disc = CRAO
- Retinal vein occlusion → retinal hemorrhage and venous engorgement

23) Thyroidectomy

- Follicular neoplasm with prior history of head/neck radiation - definitive treatment is total thyroidectomy

24) Bupivacaine

- Amides ("i" before "caine" such as bup*i*"caine) are intermediate/long acting
- Esters (no "i" before "caine" such as procaine and chloroprocaine) are short acting (tetracaine is a long-acting exception)

25) Surgical debridement

- Fournier's gangrene = extensive tissue necrosis/infection of perineum in diabetics
- Tx: urgent surgical debridement

26) Tracheostomy

- Recurrence of tumor → obstruction → difficult to maintain an endotracheal airway
- Tracheostomy would be below the lesion → maintain airway

27) Radionuclide scan with thallium and dipyridamole

- Pharmacologic stress test (dipyridamole thallium scanning - DTS); dilates coronary arteries without increasing HR or BP

28) Cyclo-oxygenase inhibition with increased norepinephrine release

- Indomethacin is an NSAID that inhibits PGE2 synthesis via COX inhibition
- Increased arterial tension is the natural way that the PDA closes

29) Ruptured diaphragm

- Traumatic disruption of hemidiaphragm → gas occupying the lower third of the left hemithorax

30) Ceftriaxone and vancomycin therapy

- No spleen → susceptible to encapsulated organisms

31) Collapse of the right middle lobe of the lung from decreased inspiratory effort

- Hasn't been out of bed yet, POD#1, egophany (consolidated or collapsed) → atelectasis

32) Intraductal papilloma

- Most common cause of nipple discharge (serous or bloody)
- Mammary duct ectasia = mass + green/brown discharge

33) Reintubation and mechanical ventilation

- Elevated PCO2 = hypoventilation
- Oxygenation is adequate (do *not* need to increase FiO2)

34) Choledocholithiasis

- Cholangiography/imaging was not performed, suggesting a retained stone in bile duct
- Normal AST/ALT rule out cirrhosis
- No weight loss/painless jaundice rule out pancreatic cancer

35) Rupture of the thoracic aorta

- "hidden injury" - asymptomatic until hematoma contained by the adventitia expands
- 1st rib break, presence of wide mediastinum, severe deceleration injury should all raise suspicion

36) Secondary hyperparathyroidism

- Hypocalcemia → ↑ PTH secretion → ↑ ALP
- CKD → ↑ creatinine and ↑ phosphate

37) Cholecystitis

- Risk factors: female, fat (BMI >30), fertile, forty

38) Mitral valve prolapse

- Mid-systolic click = MVP
- MVP syndrome is associated with nonspecific sx's (palpitations, dyspnea, dizziness, panic/anxiety attacks, numbness/tingling, etc.)

39) Torn meniscus

- Medial joint line tenderness with effusion = Meniscal; pain when getting up from a low position is a classic presentation
- Medial joint line tenderness without effusion = MCL

40) Increased serum tumor necrosis factor concentration

- Cachectic (weight loss, muscle atrophy, fatigue) is mediated by cytokines TNF, IFN- γ , IL-1, and IL-6. Appetite is usually suppressed (via hypothalamus) and BMR is increased → ↑ lipolysis, hypotriglyceridemia.

41) Basal cell carcinoma

- Upper lip, pearly surface, telangiectasia = basal cell carcinoma
- Lower lip = squamous cell carcinoma

42) Prednisone therapy and temporal artery biopsy immediately

- Requires immediate prednisone therapy to prevent blindness and bx to confirm dx

43) Apocrine glands

- Hidradenitis Suppurativa = inflammation of apocrine sweat glands

44) Observation

- Primary spontaneous pneumothorax

45) Exploratory celiotomy

- Atherosclerosis, lactic acidosis and leukocytosis = acute mesenteric ischemia
- Celiotomy (laparotomy) is needed to resect dead bowel

46) Retrograde urethrography

- High-riding prostate + blood at urethral meatus = urethral injury
- Suspected urethral injury should be evaluated with retrograde urethrogram
- IV pyelogram is useful for suspected ureter injuries; Cystoscopy is useful for suspected bladder injuries (should rule out urethral injury first, as cystoscope must go through urethra and could exacerbate existing injuries)

47) Cancel the operation

- Respect the patients decision and determine rationale for refusal

48) Varicocele

- "ropy mass" = "bag of worms"
- Stasis of venous blood → ↑ temp → damage to sperm

49) Exploratory operation

- Torsion of ovarian cyst
- Tx: surgical uncoiling and possible oophoropexy to fixate ovary (prevent recurrence)
- Drainage is contraindicated as it increases risk of spread/seeding

50) Insulinoma

- Whipple triad: low blood glucose, sx's of hypoglycemia, resolution of sx's with normalization of blood glucose
- Tx: surgical resection